Learning Spatio-Temporal Dynamic Event Voxels for Robust Event-based Object Tracking

Yuting He, Bin Fan✉, Zhongyang Ren, Qi Liu, Yuchao Dai✉

***Abstract*—Event-based object tracking has garnered significant attention due to its unique advantages in high-speed and high-dynamic-range scenarios. However, existing trackers often rely on hand-crafted event representations, which struggle to capture the complex spatio-temporal dynamics of moving objects. Although some learnable representations designed for non-tracking tasks incorporate temporal dependencies, they typically operate at a single temporal scale and lack a well-defined structure, limiting their effectiveness in the tracking domain. To this end, we propose the Spatio-Temporal Dynamic Voxel (STDV), a novel, multi-scale, and well-structured event representation that is jointly optimized with the tracking network in an end-to-end manner. Specifically, STDV first discretizes the asynchronous event stream into multi-level voxels with varying temporal resolutions. Within each level, cascaded temporal dilated convolutions are then employed to progressively expand the receptive field, capturing long-range temporal dependencies. Finally, the representations from all levels are aggregated to model tracking-adaptive dynamic context in a coarse-to-fine fashion. Furthermore, leveraging the multi-scale architecture of STDV, we introduce a Spatio-Temporal Variance Modeling (STVM) mechanism. STVM captures event fluctuations across temporal scales to highlight motion-intensive regions and guide the tracker's attention toward target-relevant dynamics. Experimental results on several RGB-event and event-only tracking benchmarks demonstrate that our method significantly enhances existing trackers, achieving new state-of-the-art performance.**

***Index Terms*—Event Camera, Object Tracking, Spatio-Temporal Dynamic Voxel, Spatio-Temporal Variance Modeling.**

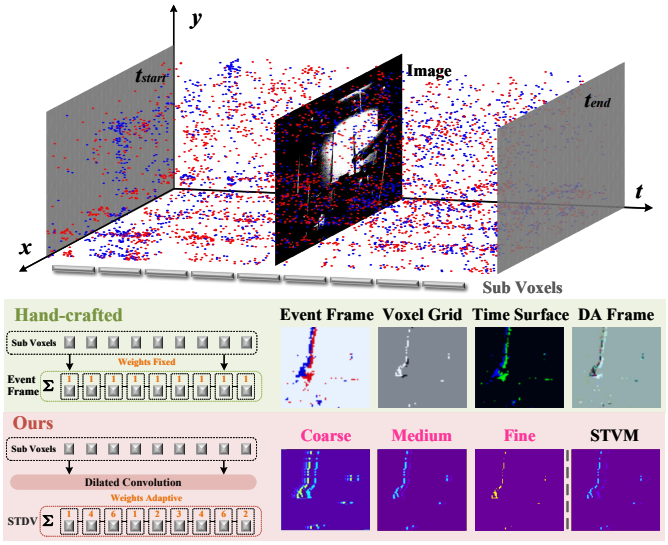


Fig. 1. **Spatio-Temporal Dynamic Voxel (STDV) *vs*. Hand-Crafted Representations.** Existing methods convert the raw events into hand-crafted representations using fixed, non-learnable aggregation weights, losing fine-grained temporal dynamics. In contrast, our STDV learns a dynamic, tracking-aware aggregation strategy. By employing a hierarchy of temporal dilated convolutions across multi-granularity voxels, STDV captures rich, coarse-to-fine spatio-temporal patterns that are crucial for robust tracking.

I. Introduction

VISUAL object tracking aims to estimate the position and scale of a given target in a video sequence [1]. This technology is a cornerstone of various fields [2], [3], [4]. Although RGB-based tracking performs well in general, its accuracy deteriorates under high-speed motion or extreme lighting conditions due to the low frame rate and limited dynamic range of conventional cameras.

Event cameras, as a type of bio-inspired asynchronous visual sensor, are revolutionizing the way visual information is acquired [5]. By responding to per-pixel changes in brightness, event cameras offer significant advantages, including low latency, high temporal resolution, high dynamic range (>120dB), and low power consumption [6]. Owing to these advantages, event-based object tracking, either leveraging event data alone or fusing it with RGB images, has emerged as a promising research field for robust target localization in challenging conditions. Among the prior works, RGB-event fusion tracking methods [7], [8], [9], [10], [11], [12] include CEUTrack [9], which employs Vision Transformers [13] for simultaneous feature extraction, fusion, and tracking; and [10], which proposes a sparsity-aware event feature extraction backbone with an adaptive bi-directional guidance module to improve tracking accuracy. Another line of work, such as HDETrack [14], represents a typical event-only tracker. It circumvents the need for additional modalities and cross-modality modules, relying solely on event data for feature extraction and tracking, thereby improving tracking speed.

Despite notable progress, most studies have focused on the design of feature extractors or cross-modal fusion modules, while the core aspect of tracking performance [9], [14], event representation, remains underexplored. Existing trackers typically convert raw events into well-structured, hand-crafted representations such as event frames [7] or voxel grids [15]. However, their reliance on fixed, non-learnable aggregation weights makes it difficult to capture complex spatio-temporal dependencies and fine-grained target dynamics (see Fig. 1). Several learnable representations [16], [17], [18] have recently

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TABLE I
COMPARISON OF EVENT REPRESENTATION METHODS. WE COMPARE THE PROPOSED STDV WITH REPRESENTATIVE HAND-CRAFTED AND LEARNABLE EVENT REPRESENTATION METHODS. H, W, AND C DENOTE THE IMAGE HEIGHT, WIDTH, AND CHANNEL DIMENSIONS, AND B IS THE NUMBER OF TEMPORAL BINS. ABBREVIATIONS: "LRN.": LEARNABLE; "T. S.": TEMPORAL SCALE; "STR.": STRUCTURED.

Representation	Dimension	Lrn.	T. S.	Str.
Event Frame [19]	$(2, H, W)$	✗	Single	High
DA Frame [20]	$(2, H, W)$	✗	Single	High
Time Surface [21]	$(2, H, W)$	✗	Single	High
Voxel Grid [22]	(B, H, W)	✗	Single	High
EST [17]	$(2, B, H, W)$	✓	Single	Low
Matrix-LSTM [18]	(C, H, W)	✓	Single	Low
NEST [16]	(B, C, H, W)	✓	Single	Low
STDV (Ours)	$(9, H, W)$	✓	Multiple	High

been proposed that implicitly model temporal dependencies via neural networks, as summarized in Tab. I. These methods can effectively capture fine-grained temporal information and perform well in tasks such as image enhancement or object recognition by preserving pixel fidelity and static semantics. However, they typically operate at a single temporal scale and lack a tracking-oriented structural design, which limits their ability to perceive diverse temporal resolutions and prevents them from explicitly modeling multi-scale temporal dependencies, dynamic contextual cues and motion-salient regions that are critical for accurately localizing targets in robust event-based tracking.

To overcome these issues, we propose an efficient and well-structured event representation method, namely Spatio-Temporal Dynamic Voxel (STDV), designed for end-to-end joint optimization with the tracking network. As a plug-and-play representation, STDV can be seamlessly integrated into any event-based tracking task, effectively enhancing the performance of existing trackers. As illustrated in Fig. 1, our core idea is to model spatio-temporal dynamics in a hierarchical and adaptive manner. STDV first partitions the event stream into multiple sub-voxels with varying temporal resolutions, creating a hierarchical view of motion at multiple temporal granularities. This helps the tracker capture both short-term motion states and long-term motion trends. Within each granularity, a cascade of temporal dilated convolutions progressively expands the temporal receptive field, enabling the model to learn a dynamic aggregation strategy that adaptively integrates event contextual information across multiple temporal scales, thus preserving motion dynamics. This hierarchical architecture offers the tracker a rich, coarse-to-fine depiction of motion dynamics while maintaining both spatial and temporal information. Crucially, our representation process maintains a well-defined structure, as it formulates the conventional event voxel as a special, single-scale instance of STDV. This principled generalization provides a clear structural underpinning for our method's design.

Building upon the dynamic and multi-scale architecture of STDV, we first obtain rich spatio-temporal motion information across multiple temporal granularities. We further observe that this information can provide a reliable cue for cross-modal fusion, guiding RGB-event interactions. Motivated by this, we introduce the Spatio-Temporal Variance Modeling (STVM) mechanism, specifically designed for cross-modal tracking. Concretely, we integrate STVM as a simple yet practical motion-saliency prior to guide motion-driven modality fusion and feature refinement. This process leverages STDV's multi-granularity structure to compute event activity variance across temporal scales, which naturally serves as an indicator of motion intensity, spatial dynamic saliency, and modality reliability. By focusing the tracker on target-relevant motion cues, STVM improves robustness in complex dynamic scenes.

In summary, our main contributions are as follows:

- We propose the Spatio-Temporal Dynamic Voxel (STDV), a novel, end-to-end learnable event representation. It dynamically captures multi-scale spatio-temporal dependencies, fully unlocking the rich dynamic information within the event stream for the tracking task, while maintaining a well-structured design.
- We introduce the Spatio-Temporal Variance Modeling (STVM) mechanism, which exploits STDV's multi-granularity structure to explicitly model motion fluctuations, thereby highlighting motion-salient regions and enhancing the tracker's adaptability to target dynamics.
- We integrate our proposed method into several state-of-the-art (SOTA) trackers and conduct comprehensive experiments on public event-based tracking benchmarks. The results demonstrate that our method yields significant performance improvements, validating the effectiveness of our dynamic event representation paradigm.

II. RELATED WORK

RGB-based Tracking. Visual object tracking is a fundamental task in computer vision. With the advancement of deep learning, RGB-based trackers have made significant progress. Early CNN-based methods like SiamFC [23] and SINT [24] pioneered feature extraction and similarity matching. Later works [25], [26], [27], [28], [29], [30] further improved feature representation and localization. Recently, Transformer-based methods [31], [32], [33], [34], [35], [36], [37] have further advanced performance. Among them, OSTrack [33] improves inference efficiency via an "early candidate elimination" module, LMTrack [35] enables flexible feature association through dynamic token selection, and CompressTracker [36] uses stage-wise distillation to enhance compactness and generalization. However, under extreme cases, *e.g.*, over-exposure or high-speed motion, these RGB-based trackers suffer significant performance degradation.

Event-based Tracking. Event camera asynchronously captures brightness changes at each pixel, offering advantages such as high dynamic range, high temporal resolution, and low power consumption [38], [39], [40]. Event-only trackers [14], [41], [42], which directly process the asynchronous event stream, can achieve robust performance in dynamic scenes. However, their localization accuracy degrades significantly when events become sparse. Consequently, RGB-event trackers [7], [8], [9], [43], [44], [45], [46], [47], [48], [49] have developed rapidly. AFNet [8] introduced cross-modal alignment

and fusion modules for collaborative, high-frame-rate tracking. CEUTrack [9] integrated feature extraction, interaction, and tracking into a single framework. OneTracker [45] unified RGB and RGB+X tracking by pretraining a foundation tracker on RGB tracking and adapting it to downstream multimodal tracking tasks via prompt-based efficient tuning. SUTrack [46] further proposed a simple unified single-object tracking framework that handles RGB-based, RGB-depth, RGB-thermal, RGB-event, and RGB-language tracking within one model and one training session. More recently, XTrack [50] explored cross-modal knowledge sharing during multimodal training. It introduced a Mixture of Modal Experts framework with a soft routing mechanism, allowing samples from one modality to exploit experts associated with other modalities when their representations are sufficiently aligned. In this manner, knowledge learned from paired RGB-event, RGB-depth, and RGB-thermal data can be shared across different sensing domains. CSTrack [51] addressed the redundant computation caused by maintaining separate RGB and auxiliary-modality feature spaces. It employed a Spatial Compact Module to integrate RGB-X inputs into a compact spatial representation and a Temporal Compact Module to construct dense temporal features from refined target-distribution maps. UETrack [52] further focused on efficient unified multimodal tracking. It supported RGB, depth, thermal, event, and language inputs within a lightweight framework and introduced a Token-Pooling-based Mixture-of-Experts module together with target-aware adaptive distillation to achieve a favorable speed–accuracy trade-off. HRCEUTrack [15] enhanced tracking via cross-modal mask modeling and orthogonal high-order regularization. TENet [10] designed an event feature extractor that utilizes an adaptive bidirectional guide fusion module to strengthen the correlation between RGB and event data. CrossEI [43] combined adaptive event sampling with bidirectional fusion to achieve robust tracking. Nevertheless, their dependence on hand-crafted event representations limits their capacity to effectively capture and model the complex spatio-temporal dynamics of moving objects, which are crucial for achieving robust and accurate performance in object tracking tasks.

Event Representation. Asynchronous event data typically need to be converted into grid-like representations to ensure compatibility with deep neural networks. Existing tracking methods predominantly leverage hand-crafted event representations. Among them, the event frame [19], [53] accumulates events within a fixed time interval to form an image-like structure. Although simple, it is sensitive to the density of events. The time surface [21] stores the latest event timestamp per pixel to achieve temporal continuity. Yet, its susceptibility to noise limits its performance in complex scenes. The voxel grid [22] divides the event stream into a sequence of temporal bins, producing an effective dense tensor representation that balances temporal resolution and robustness. Despite being well-structured, these hand-crafted representations rely on fixed aggregation weights, which limits their expressiveness. Moreover, the non-learnable nature constrains their adaptability for robust tracking. In recent years, several learnable representations have been introduced in tasks such as image enhancement [16] and object recognition [18]. A prevailing representative is the Event Spike Tensor (EST) [17], which leverages learnable kernels to convolve and sample events, providing strong temporal modeling yet limited by local temporal perception. The Neural Event STack (NEST) [16] ConvLSTM [54] is used to encode event sequences, preserving spatio-temporal information in a structured manner but incurring high computational complexity due to its inherently sequential nature. Matrix-LSTM [18] models pixel-wise event sequences using LSTM [55], providing powerful temporal dynamics for local event streams but suffering from restricted spatial receptive fields. In summary, existing representations are not explicitly tailored for the unique demands of event-based single object tracking, rendering them sub-optimal. To bridge this gap, we propose STDV, a novel, learnable, and well-structured event representation. By capturing multi-granularity spatio-temporal dependencies and enabling joint optimization with existing trackers, STDV directly addresses these prior limitations.

III. Approach

A. Overview

As shown in Fig. 2, we illustrate the general framework for event-based tracking methods, including both RGB-event trackers [7], [10] and event-only trackers [14]. For the RGB-event tracking pipeline, the key stages include: (1) Event representation, converting the raw event stream into a grid-like tensor; (2) Dual-modal feature extractor, in which backbones process RGB frames and event representations separately; (3) Cross-modal fusion, integrating features from both modalities; and (4) Tracking head, which predicts the target's state based on the fused features. Similarly, for the event-only tracker, feature extraction is performed solely on the event modality, without any cross-modal fusion. While significant efforts have been dedicated to designing sophisticated feature extraction and fusion modules, the crucial event representation stage has remained largely underexplored. Specifically, existing tracking methods [9], [10], [14] mainly rely on hand-crafted event representations. Moreover, contemporary learnable representations [16], [17], [18] lack a specific design for tracking tasks. These event representations are also limited by a single temporal scale, preventing them from adaptively capturing the full range of dynamics in the event stream. This creates a fundamental bottleneck, limiting the potential of event data in tracking.

To address this issue, we introduce innovations at two key stages of the pipeline. *First*, we revolutionize the event representation stage by proposing STDV. STDV is a learnable and tracking-aware representation that dynamically models spatio-temporal features to capture complex motion patterns. *Second*, we enhance the cross-modal fusion process of RGB-event trackers with our STVM mechanism. By leveraging the rich dynamic information from STDV, STVM explicitly models motion fluctuations and embeds this awareness into the fusion module (*i.e.*, the attention mechanism), guiding it to focus on salient dynamic regions. Notably, both STDV and STVM are plug-and-play modules that seamlessly integrate

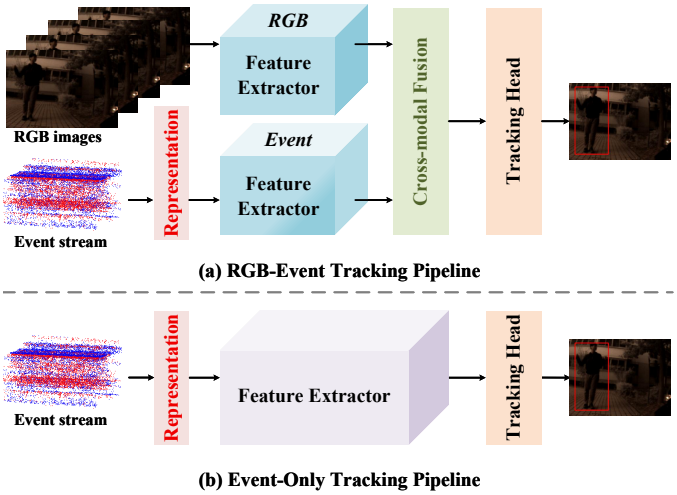


Fig. 2. **Illustration of general pipelines for event-based tracking.** (a) and (b) show the overall frameworks of RGB-event and event-only trackers, respectively. Our work introduces two key innovations: STDV, which revolutionizes the event representation stage by replacing hand-crafted event representations with a dynamic, learnable one; and STVM, which enhances the cross-modal fusion stage by injecting motion-aware guidance, enabling the model to focus on salient dynamic cues.

into various event-based trackers to boost performance, as validated in Sec. IV-D.

B. Spatio-Temporal Dynamic Voxel (STDV)

From Asynchronous Events to Multi-Granularity Voxel Grids. Our approach begins by processing the raw event stream, which is a sequence of asynchronous tuples $\mathcal{E} = \{e_i\}_{i=1}^{N}$. Specifically, each event $e_i = (x_i, y_i, t_i, p_i)$ represents a brightness change at pixel location (x_i, y_i), timestamp t_i, with polarity $p_i \in \{-1, +1\}$. To make the sparse, asynchronous event data compatible with deep neural networks, we first transform it into a grid-like representation. We adopt the widely-used voxel grid representation [22] as our foundation, leveraging its discrete and structured nature as a canvas for our dynamic temporal modeling.

Specifically, to align with the RGB frame captured over the time interval $[t_{\text{start}}, t_{\text{end}}]$, we collect all events within this window. This interval is then uniformly partitioned into B temporal bins, establishing a temporal resolution where each bin has a duration of $\Delta t = (t_{\text{end}} - t_{\text{start}})/B$. Each event e_i is then assigned to its neighboring temporal bins via a bilinear interpolation scheme. The value at a specific voxel coordinate (x, y, b) is computed by summing the contributions of relevant events based on their timestamps t_i:

$$E_b(x,y) = \sum_{i=1}^{N} p_i \max\left(0,\ 1 - \left| b - \frac{t_i - t_{\text{start}}}{t_{\text{end}} - t_{\text{start}}} (B-1) \right| \right), \quad (1)$$

where p_i denotes the event polarity and $b \in [1, B]$. We use a non-polar version where $p_i = 1$. This process yields a dense event voxel $V \in \mathbb{R}^{B \times H \times W}$.

The choice of the number of temporal bins, B, is critical as it directly dictates the temporal granularity of the representation. Nevertheless, event streams in dynamic scenes often contain a complex mixture of motions occurring at different speeds and scales, making a single, fixed choice of B inherently sub-optimal. In general, a fine granularity (large B) generates high-temporal-resolution motion representations, excelling at capturing rapid, transient movements. Yet, its sensitivity to event stream sparsity and noise can impede the learning of stable motion patterns. Conversely, a coarse granularity (small B) aggregates more events to produce smoother motion representations that effectively suppress noise and model long-term motion trends. However, this temporal aggregation may blur instantaneous motion edges, potentially compromising localization precision during abrupt movements. Thus, a clear trade-off emerges: fine granularity provides motion acuity, while coarse granularity ensures temporal robustness. A synergistic combination of both is therefore essential to yield a more complete motion representation for robust tracking in complex scenarios.

This inherent trade-off motivates our multi-granularity design. Therefore, we generate multiple voxels from the input event stream, each with a different number of bins ($B_{\text{coarse}}, B_{\text{medium}}, B_{\text{fine}}$), as shown in the green dot at the bottom row in Fig. 3. The multiple voxels designing provides the tracking model with a comprehensive, multi-scale view of the target dynamics, laying the groundwork for the hierarchical modeling performed by STDV.

Temporal Dynamic Block (TDB). Having generated multi-granularity event voxels with $B_{\text{coarse}}, B_{\text{medium}}, B_{\text{fine}}$ bins through Eq. (1), we now introduce our core building block for multi-scale temporal modeling: the Temporal Dynamic Block (TDB). A TDB is designed to operate on a single-granularity voxel and dynamically capture temporal dependencies across the full time interval of $[t_{\text{start}}, t_{\text{end}}]$.

A key challenge in processing the event voxels is to effectively model long-range temporal dependencies. A naive approach, such as using a very large kernel convolution, is computationally inefficient. To address this, we employ temporal dilated convolutions. This mechanism expands the receptive field to cover long temporal spans while maintaining a compact kernel size and manageable computational load, as illustrated in Fig. 3.

A TDB takes a single-granularity event voxel $V \in \mathbb{R}^{B \times H \times W}$ as input. For each spatial location (x, y), we treat the temporal dimension as a 1D signal of length B. The TDB then applies a stack of L 1D dilated convolutional layers along this temporal dimension, to capture dynamics across multiple temporal scales within a single pathway. Specifically, at the l-th level, the 1D convolution uses a kernel size of k and a dilation rate of $d_l = 2^{l-1}$. The number of levels, L, is determined by the number of temporal bins, B, of the input event voxel V. Our goal is to ensure the final level's receptive field spans the entire temporal duration. The effective temporal receptive field size, R_L, after L levels with a kernel size k is:

$$R_L = 1 + \sum_{l=1}^{L} 2(k-1) \cdot d_l = 1 + 2(k-1) \cdot (2^L - 1). \quad (2)$$

For each granularity level (*e.g.*, B_{coarse}, B_{medium}, B_{fine}), we select a corresponding number of levels (L_{coarse}, L_{medium}, L_{fine}) such that the condition $R_{L_i} \geq B_i$ is met. This principled

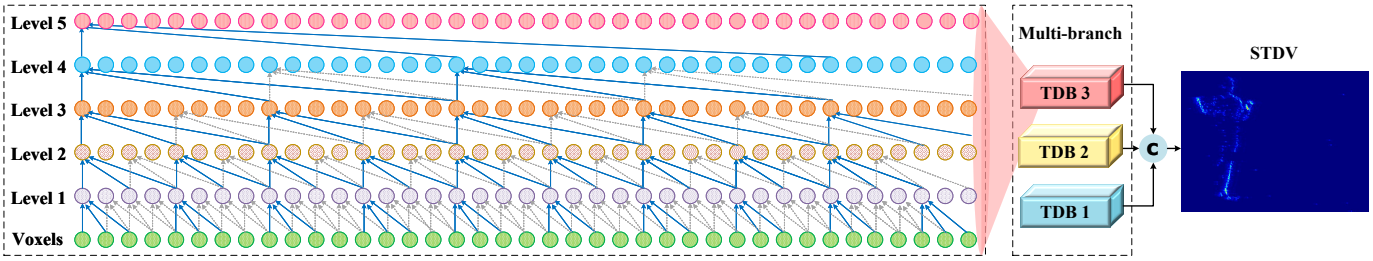


Fig. 3. **Temporal receptive field expansion via dilated convolutions.** We illustrate our temporal modeling mechanism in the Temporal Dynamic Block (TDB). We employ a stack of temporal dilated convolutions (five levels shown for visualization). The dilation rate increases at higher levels (*e.g.*, $d = 1, 2, 4, 8, 16, ...$), endowing each output element to have a growing receptive field over the input temporal bins of the event voxel (bottom row), as indicated by the blue arrows. This efficient design allows the progressive modeling of both short-term (at lower levels) and long-term (at higher levels) dependencies within a single block. Our STDV applies this principle across multiple temporal granularities to build a hierarchical representation, detailed in Fig. 4.

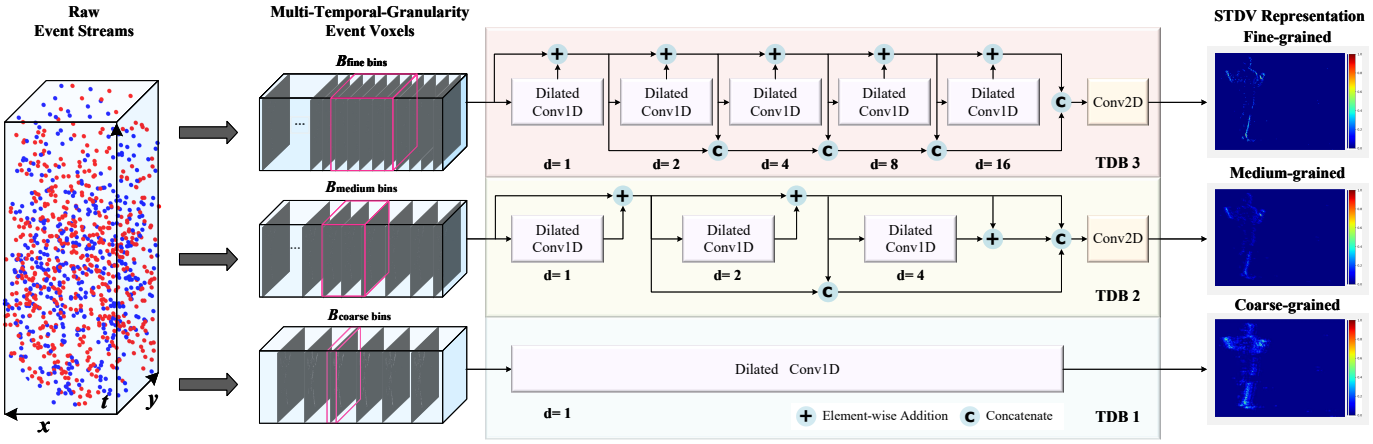


Fig. 4. **Hierarchical architecture of our STDV.** Our STDV processes the input event stream in three parallel branches, each corresponding to a different temporal granularity (Coarse, Medium, Fine). Each branch employs a TDB with a specific number of cascaded dilated convolutions to model dynamics at its scale. Within the medium and fine branches, multi-scale temporal features from all dilated convolutions are first concatenated, then spatially aggregated via a 2D convolution. Finally, the feature maps from all three branches are concatenated to form the final STDV representation.

design ensures that for every granularity, our TDB can model dependencies across the entire time window.

A critical aspect of our TDB is that we leverage the output of *every* dilated convolutional layer. The output of the l-th level, F_l, encapsulates temporal dynamics at a specific scale corresponding to its receptive field R_l. By collecting the feature maps from all levels $F_1, F_2, \ldots, F_L$, the TDB produces a rich set of multi-scale temporal features. These features model the target's motion dynamics and variations from coarse to fine, which helps enhance the temporal consistency of tracking. This process is then applied independently at each spatial location, preserving the spatio-temporal structure for the subsequent stage. It is worth noting that *existing hand-crafted event representations can be viewed as degenerate cases of our TDB*. For instance, an event frame is analogous to a TDB with a single level and fixed, uniform weights (*i.e.*, uniform counting), while a standard voxel grid corresponds to a TDB with fixed kernels performing predefined linear interpolation. This principled generalization shows a structural and interpretable foundation for our STDV representation.

Hierarchical STDV Architecture. We assemble our complete STDV by integrating multiple TDBs in a hierarchical fashion. As depicted in Fig. 4, in addition to perceiving the multi-scale temporal context in each TDB, STDV also concurrently processes event data at multiple temporal granularities (*i.e.*, B_{coarse}, B_{medium}, and B_{fine}) to achieve a comprehensive understanding of motion patterns.

The three branches of STDV operate in parallel, each tailored to a specific temporal granularity. Let's take the "Medium-grained" branch as an example. It first takes an event voxel V_{medium} with $B_{\text{medium}} = 30$ bins as input. This voxel is then processed by a TDB composed of $L_{\text{medium}} = 3$ cascaded temporal dilated convolutions (with dilations of 1, 2, and 4). Finally, the TDB leverages the outputs from all its levels, which are concatenated and then passed through a 2D convolutional block, yielding $F_{\text{medium}} \in \mathbb{R}^{L_{\text{medium}} \times H \times W}$. This fuses the multi-scale features and models local spatial patterns for the medium granularity. The "Coarse-grained" ($B_{\text{coarse}} = 6$) and "Fine-grained" ($B_{\text{fine}} = 126$) branches follow the same principle, employing TDBs with $L_{\text{coarse}} = 1$ and $L_{\text{fine}} = 5$ levels. The number of levels in each branch is deliberately chosen by referring to Eq. (2), such that its temporal receptive field can perceive long-range dependencies within the corresponding granularity.

The entire process can be formulated as:

$$F_g = \begin{cases} [C_l^{1D}(V_g)]_{l=1}^{L_g}, & g = \text{coarse}, \\ C^{2D}\big([C_l^{1D}(V_g)]_{l=1}^{L_g}\big), & g \in \{\text{medium}, \text{fine}\}, \end{cases} \tag{3}$$

where V_g is the input event voxel, $C_l^{1D}(\cdot)$ is the l-th temporal dilated 1D convolution, $[\cdot]_{l=1}^{L_g}$ denotes concatenation along channels, $C^{2D}(\cdot)$ is a 2D convolution, and F_g is the resulting feature map. Therefore, the feature maps $F_{\text{coarse}}, F_{\text{medium}}, F_{\text{fine}}$, produced by the three parallel branches, individually capture target dynamics at varying temporal granularities. Finally, these maps are then concatenated along the channel dimension, yielding the final STDV representation $F_{\text{STDV}} \in \mathbb{R}^{L' \times H \times W}$, where $L' = L_{\text{coarse}} + L_{\text{medium}} + L_{\text{fine}} = 9$. Such an expressive, multi-granularity tensor effectively provides a comprehensive and robust view of the target's motion state, which is then fed into the subsequent stages of the tracking network.

C. Spatio-Temporal Variance Modeling (STVM)

In RGB-event object tracking, modality reliability dynamically changes with the scene, posing challenges for effectively guiding cross-modal attention. Although our STDV representation effectively captures *how* a target moves, fully leveraging this information still requires addressing these challenges. To address this issue, we propose explicitly modeling the intensity of target motion and using it as a motion-saliency prior to guide the cross-modal attention. Therefore, we further introduce the Spatio-Temporal Variance Modeling (STVM) mechanism for enhancing the tracker's robustness. The key insight behind STVM is that the variance across the multi-granularity event representations produced by STDV can serve as a potent proxy for motion intensity. A high variance at a specific spatial location signals significant dynamic activity, as the motion patterns differ across various time scales.

To this end, we compute the variance of F_{STDV} along its channel dimension for each spatial location (x, y):

$$M_{\text{var}}(x, y) = \text{Var}(F_{\text{STDV}}(:, x, y)). \tag{4}$$

This operation yields a 2D variance map $M_{\text{var}} \in \mathbb{R}^{H \times W}$, which effectively acts as a motion saliency map. To serve as a stable guidance signal, this map is then normalized to a range of $[0, 1]$ using min-max normalization:

$$\hat{M}_{\text{var}} = \frac{M_{\text{var}} - \min(M_{\text{var}})}{\max(M_{\text{var}}) - \min(M_{\text{var}}) + \epsilon}, \tag{5}$$

where ϵ is a small constant set to 1×10^{-6} to prevent division by zero.

STVM-Guided Cross-Modal Attention. The normalized variance map $\hat{M}_{\text{var}}$ provides an explicit estimate of motion saliency across spatial locations. To exploit this signal during multimodal fusion, we inject it into the Cross-Modal Fusion module to guide the attention computation. Specifically, STVM leverages the multi-scale motion information extracted from STDV as a motion-saliency prior, guiding attention toward dynamically active regions and enhancing RGB-event feature interaction (as illustrated in Fig. 5). Since STVM only acts on the intermediate RGB-event interaction process, it can be readily adapted to different RGB-event tracking frameworks. In our implementation, STVM is applied at the cross-modal fusion stage, after RGB and event feature extraction and before the tracking head, rather than inside the backbone. It is implemented as a single attention modulation

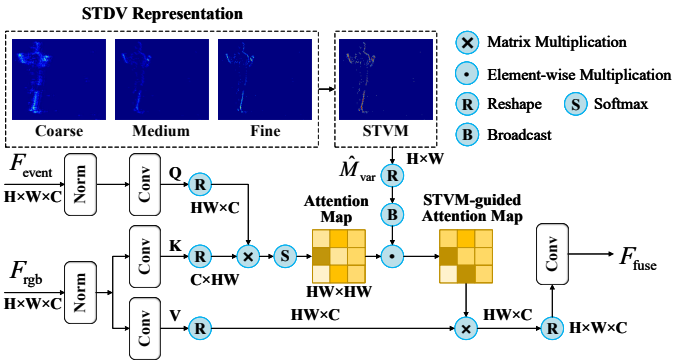


Fig. 5. **Illustration of the STVM-guided attention.** Our STVM mechanism enhances the standard cross-modal attention by injecting motion-aware guidance. The normalized variance map $\hat{M}_{\text{var}}$, computed from the STDV representations, acts as a spatial weight. It is multiplied element-wise with the standard attention map, effectively compelling the model to focus on the dynamically active regions with high motion intensity. This process yields a motion-aware fused feature that is more robust against background distractors.

module, while for multi-scale fusion architectures, $\hat{M}_{\text{var}}$ can be resized and independently applied to each selected feature scale. We then adopt a standard scaled dot-product attention mechanism [56] as the baseline fusion strategy. Specifically, the query Q is projected from the event features, while the key K and value V are projected from the RGB features. The attention map is then computed as:

$$A_{\text{standard}} = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \tag{6}$$

where d_k is the feature dimension. To incorporate motion intensity information, the normalized variance map $\hat{M}_{\text{var}} \in \mathbb{R}^{H \times W}$ is first reshaped into a vector of size $1 \times HW$. This vector is then broadcast along the query dimension to match the spatial structure of the attention matrix. The motion-aware attention map is then computed through element-wise modulation:

$$A_{\text{STVM}} = A_{\text{standard}} \odot \text{Broadcast}(\hat{M}_{\text{var}}), \tag{7}$$

where $\odot$ denotes element-wise multiplication. This modulation explicitly biases the attention mechanism toward regions with high dynamic activity. As a result, keys located in motion-salient regions receive amplified attention weights, while static or background regions are relatively suppressed. Finally, the motion-aware fused representation is obtained as:

$$\mathcal{F}_{\text{fuse}} = \text{FFN}(A_{\text{STVM}} \cdot V), \tag{8}$$

where $\text{FFN}(\cdot)$ denotes a feed-forward network. Residual connections and layer normalization follow the standard Transformer design but are omitted here for clarity. It compels the network to pay more attention to regions with high motion activity during RGB-event feature integration.

IV. Experiment

We conduct comprehensive experiments of our method on several event-based tracking datasets, including FE240 [7] and COESOT [9] for RGB-event tracking, and EventVOT [14] for event-only tracking. Following the standard protocols as [33],

[57], [58], we adopt three widely-used metrics for performance evaluation: Success Rate (SR) [57], Precision Rate (PR) [57], and Normalized Precision Rate (NPR) [58].

A. Implementation Details

Datasets. In this paper, we conduct experiments on several high-quality event-based visual tracking datasets, including COESOT [9], FE240 [7], and EventVOT [14], which provide diverse scenes, object categories, and challenging motion and illumination conditions for comprehensive evaluation of tracking algorithms. *Specifically,*

(i) The COESOT dataset [9] is a comprehensive RGB-event tracking dataset with 1,354 video sequences covering 90 indoor and outdoor object categories. Captured with a DVS346 camera with a zoom lens, it supports scale and depth variations. Each frame is annotated with 17 attributes (such as *fast motion*, *illumination variation*) for evaluating tracking algorithms under challenging conditions. The dataset includes 827 training and 527 test videos, making it one of the most extensive publicly available datasets for RGB-event tracking.

(ii) The FE240 dataset [7] is an extended version of the earlier FE108 dataset [7], offering higher temporal resolution with annotation frequencies up to 240 Hz. The dataset contains over 143K images along with corresponding recorded events, providing a rich resource for event-based vision research. Collected using a DVS346 grayscale event camera, the dataset includes 71 training and 25 test videos, with over 1.13 million annotations. It covers various degraded conditions, making it a comprehensive resource for event-based vision research.

(iii) The EventVOT dataset [14] is the first high-resolution (1280×720) benchmark for event-based visual object tracking, designed to evaluate event cameras in complex scenarios. It contains 1,141 sequences (841 for training, 18 for validation, and 282 for testing), covering 19 target categories. This dataset includes diverse day and night scenes, highlighting event cameras' advantages in *fast motion*, *low light*, *low latency*, and *low power*. It provides a high-quality benchmark for research on high-resolution event-driven tracking.

Metrics. We adopt three standard metrics, SR, PR, and NPR, to evaluate tracking accuracy and stability. *Specifically,*

(i) SR metric [57]. The success rate represents the ratio of successfully tracked frames to the total number of frames at a given Intersection over Union (IoU) threshold. It measures tracking accuracy by evaluating the overlap between predicted and ground truth bounding boxes. A *higher* success rate indicates *better* accuracy and consistency, making it crucial for assessing a tracker's ability to maintain target stability.

(ii) PR metric [57]. The precision rate measures the distance between the tracker's predicted position and the ground truth, calculated from the pixel distance between their centers. It can be affected by target size and resolution, as small deviations may significantly impact precision.

(iii) NPR metric [58]. To overcome the limitations of precision rate, normalized precision rate (NPR) standardizes precision by adjusting for target scale and image resolution. This process converts pixel errors into a unified metric, enabling fairer comparisons of tracking performance across different target sizes and conditions.

Module Integration. To enhance the tracking performance, we integrate our proposed modules into existing advanced trackers. Notably, STDV and STVM are integrated as plug-and-play modules, where STDV replaces the original event representation and STVM is incorporated into attention-based Cross-modal Fusion modules, as illustrated in Fig. 2. Specifically, we replace the original event representation with STDV by modifying the input channels of the event feature extractor to 9. STVM is inserted at the cross-modal fusion stage, after RGB and event feature extraction and before the tracking head, rather than inside the backbone. For different fusion structures, STVM resizes the motion weight map via interpolation to match the corresponding feature scale, and incorporates it into the cross-modal attention matrix as a motion-saliency prior at the selected fusion position (see Fig. 5).

Training Details. All experiments are conducted on a workstation equipped with a single RTX 3090 GPU, running on Ubuntu 22.04.3 LTS. The implementation is based on the PyTorch 1.12.1 framework under a Python 3.7 environment. For training, we employ the AdamW optimizer with an initial learning rate of 1×10^{-4}, which decays by 10% every 48 epochs. The batch size is fixed at 20 to balance computational efficiency and GPU memory usage. We train our model on the training splits of COESOT [9], FE240 [7], and EventVOT [14] datasets, and evaluate its tracking performance on their respective test sets. *Notably, to ensure a fair comparison, all methods are retrained on the same dataset and evaluated under a unified protocol.* The input configurations are standardized as follows: template images are resized to $128 \times 128 \times 3$, while search regions are fixed at $256 \times 256 \times 3$ to ensure consistent input scales across different tasks.

B. Comparison with State-of-the-art Trackers

Results on FE240 Dataset [7]. We conduct a comprehensive evaluation on this dataset, and the experimental results are shown in Tab. II, where our method achieves the best performance. Specifically, compared to TENet [10], our method improves SR by 3.2%; compared to CrossEI [43], it achieves improvements of 4.6% in SR. These results fully validate the effectiveness of our proposed event representation and variance modeling strategy on asynchronous and sparse event data, highlighting their potential in event-driven vision applications.

Results on COESOT Dataset [9]. To evaluate the effectiveness of our method, we further conducted experiments on the COESOT dataset. Tab. III shows superior results across multiple metrics. Our method outperforms TENet [10] with a 0.5% improvement in SR, and enhances PR by 5.1% compared to CSAM [11]. These results confirm that our method exhibits strong capabilities in both accuracy and robustness.

Furthermore, our method achieves 58.1 *fps* on a single RTX 3090 GPU. Compared with SpikeFET [60], our method improves the SR by 0.4% while achieving nearly 3× higher inference speed, demonstrating that the proposed STDV and STVM modules can effectively enhance tracking performance with limited computational overhead. Overall, our method achieves a favorable balance between tracking accuracy and efficiency, providing a competitive solution for practical event-based tracking applications.

TABLE II
PERFORMANCE COMPARISON ON THE FE240 DATASET [7]. "OURS" IS BUILT UPON TENET [10] BY REPLACING THE EVENT REPRESENTATION WITH STDV AND INTEGRATING STVM INTO THE CROSS-MODAL FUSION MODULE (*cf*., FIG. 2 (A)). THE BEST RESULTS ARE SHOWN IN **BOLD**.

Tracker	TransT [32]	DeT [59]	CEUTrack [9]	FENet [7]	AFNet [8]	CrossEI [43]	TENet [10]	Ours
SR/PR (%)	49.3/76.2	54.2/81.2	55.6/84.4	55.6/84.3	58.4/87.0	59.4/92.3	60.8/91.9	**64.0/94.6**

TABLE III
PERFORMANCE COMPARISON ON THE COESOT DATASET [9]. "EF" REFERS TO EVENT FRAME, AND "EVOX" REPRESENTS EVENT VOXEL. "OURS" IS BUILT UPON TENET BY USING STDV FOR EVENT REPRESENTATIONS AND INTEGRATING STVM INTO THE CROSS-MODAL FUSION MODULE.

Tracker	Representation	SR	PR	NPR	FPS
Ours	**RGB + STDV**	**68.9**	**81.8**	**80.7**	58.1
SpikeFET [60]	RGB + EF	68.5	81.7	80.6	19.6
TENet [10]	RGB + EF	68.4	81.6	80.5	59.2
CSAM [11]	RGB + EVox	68.1	76.7	74.8	-
CMDTrack [61]	RGB + EF	65.7	74.8	-	-
ViPT [62]	RGB + EF	65.7	73.9	72.2	43.7
HRCEUTrack [15]	RGB + EVox	63.2	77.5	76.5	85.2
CEUTrack [9]	RGB + EVox	62.7	76.0	74.9	66.7
TransT [32]	RGB + EF	60.5	72.4	71.9	44.5
TrDiMP [63]	RGB + EF	60.1	72.2	71.7	23.1
ToMP101 [64]	RGB + EF	59.9	71.6	71.3	17.8
TrSiam [63]	RGB + EF	59.7	71.4	71.0	31.1
OSTrack [33]	RGB + EF	59.0	70.7	70.5	93.4
AiATrack [65]	RGB + EF	59.0	72.4	71.4	33.8
KYS [66]	RGB + EF	58.6	71.6	71.3	17.8
PrDiMP50 [29]	RGB + EF	57.9	69.6	69.4	26.7
STARK-ST50 [67]	RGB + EF	56.0	67.7	67.2	37.4
MixFormer1k [31]	RGB + EF	56.0	66.6	66.3	22.2
ATOM [27]	RGB + EF	55.0	68.8	68.3	26.7

TABLE IV
PERFORMANCE COMPARISON OF EVENT-ONLY TRACKERS ON THE EVENTVOT DATASET [14]. "OURS" IS BUILT UPON HDETRACK BY REPLACING THE EVENT REPRESENTATION WITH STDV.

Tracker	OSTrack [33]	SimTrack [68]	PrDiMP [29]
SR/PR	54.3/56.8	55.4/54.2	55.5/57.2
Tracker	MambaEVT-P [42]	HDETrack [14]	Ours
SR/PR	55.4/55.9	56.0/58.8	**56.7/60.3**

Results on EventVOT Dataset [14]. We also conduct experiments on the EventVOT dataset, a purely event-based benchmark, to further validate the effectiveness of our method. Tab. IV reports the experimental results. We can see that simply replacing the original representation in HDETrack [14] with our STDV leads to performance improvements, increasing the SR metric by 0.7%. This result thoroughly validates the significant potential and plug-and-play nature of our event representation by the proposed STDV for event-only tracking.

Efficiency Evaluation. To evaluate the computational complexity and inference speed of our method, we conduct tests based on the TENet [10] under different input configurations. Tab. V reports the performance metrics of different event representations in terms of "MACs", "Parameters", and "Tracking Speed". Experimental results show that our method incurs a computational cost of 37.546 G, which represents only a marginal increase of 2.389 G compared to TENet [10]. Importantly, the model's parameters remain nearly unchanged, indicating that the performance gains come with virtually no additional parameter overhead. When tested on a single RTX 3090 GPU, our method achieves 58.1 *fps*. Moreover, a significant performance improvement is observed, with SR metric increasing by 3.2% (see Tab. II).

TABLE V
EFFICIENCY EVALUATION OF MODEL COMPLEXITY AND SPEED. ↓ MEANS LOWER IS BETTER, AND ↑ MEANS HIGHER IS BETTER. THESE REPRESENTATIONS HAVE BEEN DEPLOYED ONTO THE TENET [10].

Representations	MACs ↓	Params ↓	FPS ↑
RGB & Event Frame [19]	35.157 G	130.319 M	59.2 *fps*
RGB & EST [17]	37.327 G	130.320 M	58.7 *fps*
RGB & NEST [16]	415.325 G	131.101 M	12.0 *fps*
RGB & Matrix-LSTM [18]	38.274 G	130.324 M	53.7 *fps*
RGB & **STDV (Ours)**	37.546 G	130.322 M	58.1 *fps*

TABLE VI
ABLATION STUDIES ON THE FE240 DATASET [7]. HERE, STVM HAS NOT BEEN INTEGRATED.

	Methods	SR	PR	NPR
Diverse Event Representations	01) RGB & Event Frame [19]	60.8	91.9	65.1
	02) RGB & Event DA Frame [20]	61.5	89.0	67.7
	03) RGB & Event TimeSurface [21]	60.9	87.9	67.8
	04) RGB & Voxel Grid [22]	61.2	90.7	66.7
	05) RGB & EST [17]	31.7	50.4	31.9
	06) RGB & NEST [16]	53.4	82.1	55.9
	07) RGB & Matrix-LSTM [18]	54.0	81.3	58.2
	08) RGB & DES [69]	62.6	90.9	67.6
	09) RGB & TMA [70]	63.0	92.6	67.9
	10) RGB & STDV	**63.3**	**93.2**	**69.1**
Multi-Granularity Event Voxels	11) 6bins	60.9	90.9	65.7
	12) 30bins	61.2	90.7	66.7
	13) 126bins	61.6	92.0	66.7
	14) 6bins & 30bins	61.6	91.4	66.6
	15) 6bins & 126bins	61.3	90.8	66.7
	16) 30bins & 126bins	60.0	90.0	64.2
	17) 6bins & 30bins & 126bins	59.8	88.6	65.4
Diverse Temporal Granularities	18) TDB 1	61.8	92.0	67.7
	19) TDB 2	61.8	91.0	67.5
	20) TDB 3	62.0	91.9	67.9
	21) TDB 1 & TDB 2	62.7	92.1	68.6
	22) TDB 1 & TDB 3	62.5	92.6	68.2
	23) TDB 2 & TDB 3	62.9	92.6	69.0
	24) TDB 1 & TDB 2 & TDB 3	**63.3**	**93.2**	**69.1**

C. Ablation Studies

Ablations on Event Representations. We conduct ablation studies on different event representations combined with RGB inputs using the FE240 dataset [7], as shown in Tab. VI (top). Traditional hand-crafted representations provide stable baselines, among which the combination of RGB and voxel grid achieves 61.2% SR, 90.7% PR, and 66.7% NPR. In

TABLE VII
EFFECTIVENESS EVALUATION OF OUR STDV AND STVM ON VARIOUS RGB-EVENT TRACKERS ON THE FE240 DATASET [7].

Tracker	Representation	SR	PR
CEUTrack [9]	Event Voxel	55.6	84.4
	STDV	58.5	89.9
	STDV & STVM	**59.4**	**90.5**
HRCEUTrack [15]	Event Voxel	47.7	73.0
	STDV	58.8	88.6
	STDV & STVM	**60.7**	**93.1**
ViPT [62]	Event Frame	59.8	86.3
	STDV	60.3	88.7
	STDV & STVM	**61.5**	**90.5**
TENet [10]	Event Frame	60.8	91.9
	STDV	63.3	93.2
	STDV & STVM	**64.0**	**94.6**

contrast, the proposed STDV consistently outperforms these methods. Moreover, existing learnable representations perform notably worse, mainly due to the mismatch between their design objectives and the requirements of the tracking task. Specifically, NEST [16] focuses on texture reconstruction for image enhancement, which may introduce redundant information, and its sequential structure leads to extremely high computational cost. Matrix-LSTM [18], while used for modeling pixel-wise dynamics, suffers from a restricted spatial receptive field, making it difficult to capture the robust spatial dependencies required for tracking tasks. Overall, these results demonstrate that STDV is a tracking-oriented event representation that dynamically captures both short-term motion states and long-term motion trends, outperforming hand-crafted and learnable approaches.

Ablations on Multi-Granularity Voxels. We further investigate whether naively concatenating event voxels with different temporal bins could improve performance. As shown in Tab. VI (middle), using a single granularity (*e.g.*, 6, 30, or 126 bins) achieves stable results, while simply combining multiple bins (*e.g.*, 6 & 30, or 6 & 30 & 126) may lead to performance degradation. This phenomenon indicates that naively concatenating single-scale features from different granularities fails to effectively model cross-scale correlations and may even introduce redundant or conflicting information. This further motivates our TDB design: before fusion, cascaded dilated convolutions first extract complementary multi-scale contexts within each granularity, thereby suppressing redundancy and enabling more efficient multi-granularity collaboration.

Ablations on TDB Branches. We evaluate the impact of different TDB configurations. As shown in Tab. VI (bottom), even a single TDB branch effectively captures multi-scale temporal dynamics at a specific granularity by using cascaded dilated convolutions (*e.g.*, 61.8% *vs.* 60.9% in SR). Notably, combining multiple TDB branches leads to a significant performance improvement. Specifically, integrating TDB 1 and TDB 2 boosts the SR to 62.7%. When all three TDB branches are combined, SR further increases to 63.3%. This multi-granularity collaboration enables STDV to comprehensively capture both short-term motion states and long-term trends, thereby enhancing tracking accuracy in complex scenes.

TABLE VIII
MOTION PRIOR COMPARISON ON THE FE240 DATASET [7].

Tracker	Motion Prior	SR	PR
TENet [10]	Count Map	62.6	92.0
	Gradient Magnitude	62.0	91.4
	STVM	**64.0**	**94.6**

TABLE IX
EFFECTIVENESS EVALUATION OF OUR STDV AND STVM ON VARIOUS RGB-EVENT TRACKERS ON THE COESOT DATASET [9].

Tracker	Representation	SR	PR
CEUTrack [9]	Event Voxel	62.7	76.0
	Ours	**63.3**	**76.2**
ViPT [62]	Event Frame	65.7	73.9
	Ours	**66.4**	**79.3**
TENet [10]	Event Frame	68.4	81.6
	Ours	**68.9**	**81.8**

D. Effectiveness Evaluation of STDV and STVM

In this section, we evaluate STDV and STVM from both quantitative and qualitative perspectives, demonstrating their effectiveness, generality, and plug-and-play adaptability across *RGB-event* and *event-only* tracking paradigms.

Quantitative Analysis. To validate the generality and effectiveness of our STDV representation and STVM mechanism, we conduct comprehensive evaluations across different tracking frameworks and datasets, including *RGB-event trackers* on FE240 dataset [7] and COESOT dataset [9], as well as *event-only trackers* on EventVOT dataset [14].

(***i***) For *RGB-event tracking* on the FE240 dataset [7], as shown in Tab. VII, simply replacing the original event representations of the four advanced RGB-event trackers with our STDV yields significant performance improvements. This demonstrates that STDV can provide a more informative event representation by preserving fine-grained spatio-temporal dynamics. Moreover, integrating STVM further improves tracking performance, demonstrating the effectiveness of the proposed motion-saliency prior. As shown in Tab. VIII, STVM achieves the best performance among different motion priors, outperforming *Count Map* by 1.4% in SR and 2.6% in PR, and *Gradient Magnitude* by 2.0% in SR and 3.2% in PR. These results suggest that STVM captures multi-granularity temporal variations more effectively than simple handcrafted priors, enabling cross-modal attention to focus on motion-salient and target-relevant regions.

For RGB-event tracking on the COESOT dataset [9], as shown in Tab. IX, replacing the original event representations of several representative RGB-event trackers with the proposed STDV representation and STVM mechanism also leads to consistent gains. Specifically, CEUTrack achieves improvements of 0.6% in SR and 0.2% in PR, while ViPT obtains a 5.4% improvement in PR; TENet also shows gains across multiple evaluation metrics. These results demonstrate that STDV and STVM are not limited to a specific tracker or dataset, but can serve as a general plug-and-play event modeling module for diverse RGB-event tracking frameworks.

It can be observed that the gains on COESOT are relatively

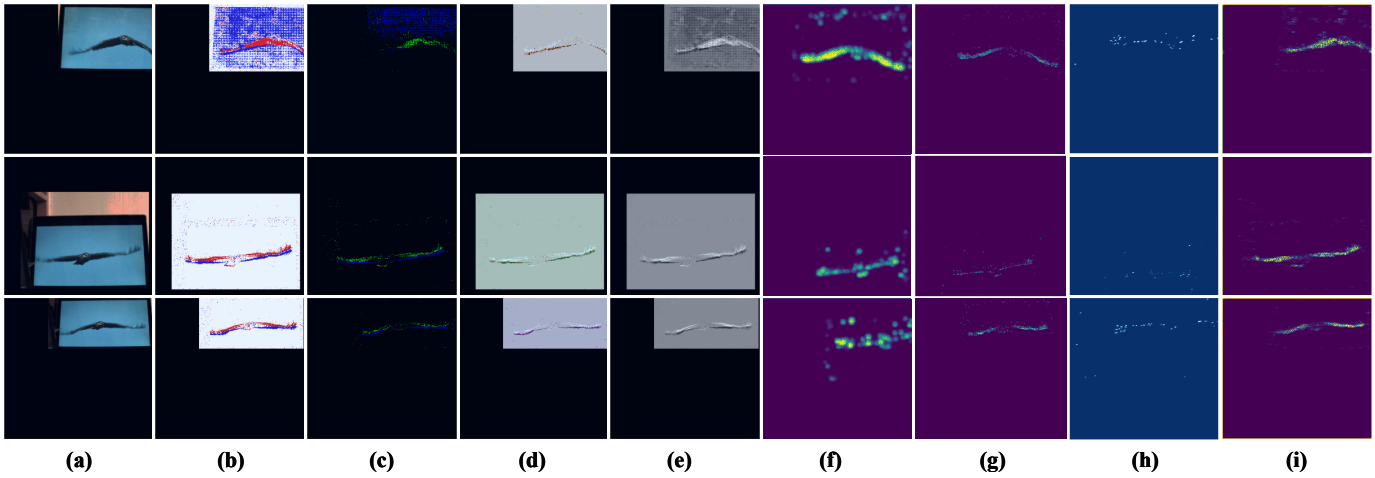


Fig. 6. **Qualitative comparison of event representations.** From left to right: (a) RGB Image, (b) Event Frame [19], (c) Event Time Surface [21], (d) Event DA Frame [20], (e) Event Voxel Grid [22], (f) Event Spike Tensor (EST) [17], (g) Neural Event STack (NEST) [16], (h) Matrix-LSTM [18] and (i) Our STDV. While existing event representations suffer from noise, sparsity, or fragmented structures, our STDV learns to produce a clean, structured, and comprehensive representation that clearly highlights the target's motion and shape.

TABLE X
EFFECTIVENESS EVALUATION OF OUR STDV ON ADVANCED EVENT-ONLY TRACKERS ON THE EVENTVOT DATASET [14].

Tracker	SR	PR	NPR
HDETrack [14]	56.0	58.8	71.4
HDETrack + STDV	**56.7**	**60.3**	**72.4**
OSTrack [33]	54.3	56.8	69.5
OSTrack + STDV	**56.4**	**58.8**	**70.8**

modest compared with those on FE240. This is mainly related to the characteristics of COESOT. Compared with FE240, COESOT contains richer RGB appearance cues and more visually distinguishable targets, which already provide strong discriminative information for RGB-event trackers. Moreover, some slow-motion sequences in COESOT provide relatively sparse event responses, which limits the contribution of the STVM-based motion prior. In contrast, FE240 contains more motion-dominated scenarios, such as fast motion, low illumination, and high dynamic range, where event information plays a more critical role. Therefore, STDV and STVM bring more pronounced improvements on FE240 by better capturing multi-scale motion dynamics and guiding cross-modal fusion toward motion-salient regions.

(***ii***) For *event-only tracking* on the EventVOT dataset [14], as shown in Tab. X, substituting the original event representations with STDV consistently improves the performance of two representative trackers. These results further demonstrate the cross-model adaptability of STDV in event-only tracking scenarios. Together, these results demonstrate that the proposed STDV and STVM exhibit strong plug-and-play capability and cross-model adaptability, effectively enhancing diverse event-based tracking paradigms.

Qualitative Visualization. To provide intuitive insights into the effectiveness of our method, we conduct qualitative analyses from three perspectives: event representation quality, tracking performance, and model response behavior.

(***i***) First, we visualize different event representations to demonstrate the advantages of the proposed STDV. As shown in Fig. 6, we compare STDV with several representative event representation methods, including widely used hand-crafted representations such as Event Frame [19], Time Surface [21], DA Frame [20], and Voxel Grid [22], as well as recent learning-based methods such as EST [17], NEST [16], and Matrix-LSTM [18]. *The hand-crafted representations* exhibit clear limitations. Event Frame is easily affected by noise, resulting in blurred object structures. Time Surface and DA Frame generate cleaner responses, but they are usually too sparse to preserve complete spatial details, as they mainly emphasize the most recent or most active events. Voxel Grid retains richer temporal information, but its fixed aggregation granularity still leads to fragmented object structures. These artifacts mainly arise from their static and non-learnable aggregation mechanisms. *Learning-based representations* alleviate these problems to some extent. NEST [16] encodes event sequences with a bidirectional ConvLSTM [54], which helps preserve spatiotemporal information and suppress noise. However, since it is originally designed for image enhancement, it tends to produce overly smooth and texture-rich representations, leading to blurred motion boundaries. Matrix-LSTM [18] models the event sequence of each pixel independently with LSTM, enabling fine-grained temporal modeling. Nevertheless, its limited spatial receptive field caused by the 1×1 convolution restricts spatial interactions, producing sparse and noise-sensitive activations that fail to form coherent foreground structures.

In contrast, the proposed STDV generates more complete and informative visual representations. By dynamically learning event aggregation across multiple temporal scales under task-aware supervision, STDV effectively suppresses noise while preserving fine-grained motion cues. As a result, the target structure and motion trajectory are more clearly delineated, demonstrating the superiority of the dynamic representation paradigm for capturing complex motion patterns in robust single object tracking.

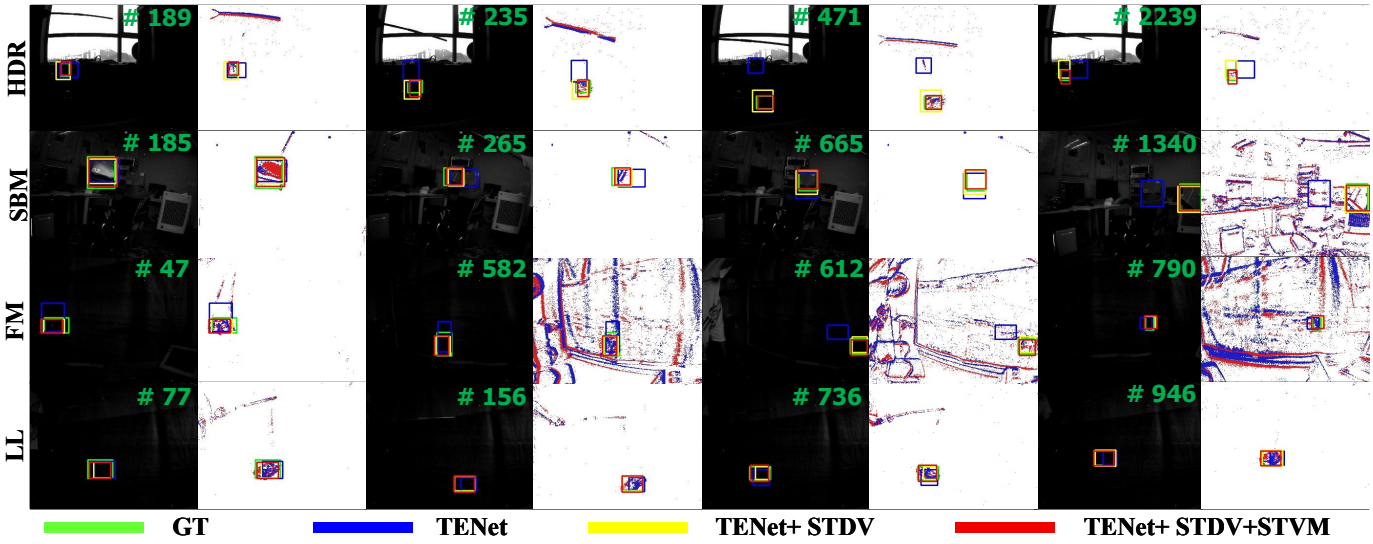


Fig. 7. **Qualitative comparison results** of our method with TENet [10] on the FE240 dataset [7]. Event images are used for visual comparison. The examples cover several challenging scenarios, including high dynamic range (HDR), low light (LL), fast motion (FM), and severe background motion (SBM). The # in the upper-right corner of each image denotes the current frame index.

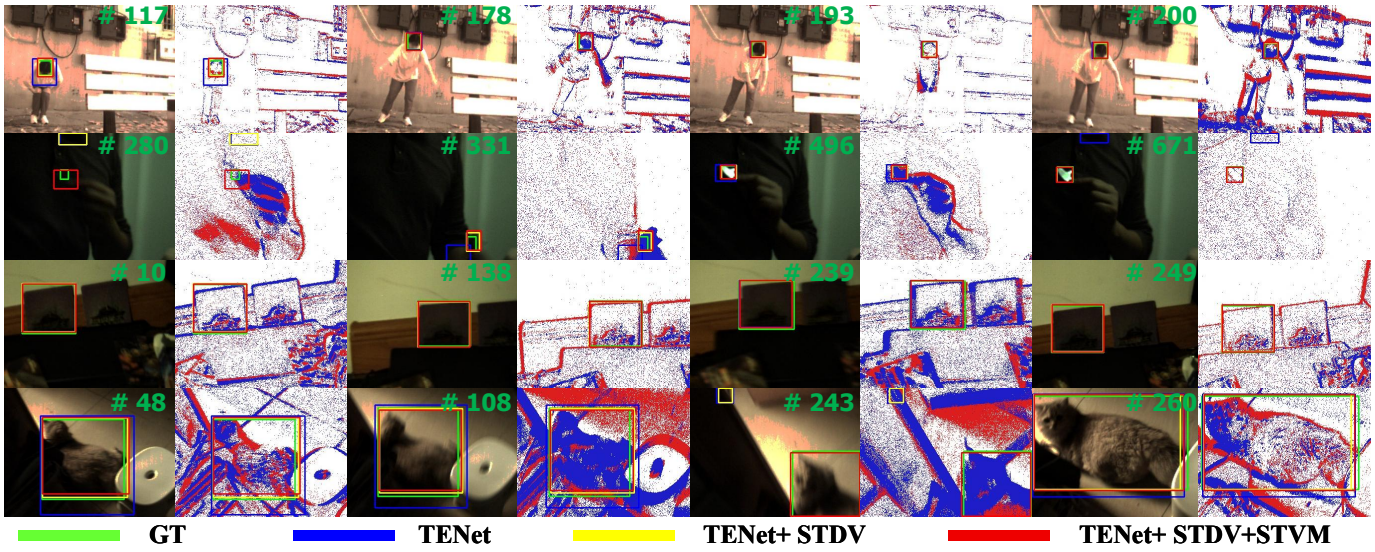


Fig. 8. **Qualitative comparison results** of our method with TENet [10] on the COESOT [9] datasets. The # in the upper-right corner of the images denotes the current frame index.

(***ii***) Second, we compare tracking results on multiple datasets. As shown in Fig. 7 and Fig. 8, our method achieves more stable and accurate localization in challenging RGB-event scenarios, including high dynamic range, low light, fast motion, severe background motion, and over-exposure. In addition, Fig. 9 presents visual results on the *event-only* EventVOT dataset, where STVM is not introduced and the evaluation focuses on STDV. These comparisons indicate that STDV provides clearer motion structures and improves tracking robustness across different datasets, while STVM further enhances localization accuracy and stability in RGB-event tracking frameworks.

(***iii***) Third, we analyze the effect of STVM through motion prior visualization. As shown in Fig. 10, compared with the diffuse activations produced by the standard attention mechanism, STVM generates more concentrated responses that are well aligned with the target region. This demonstrates that STVM can effectively suppress background distractions and provide a precise motion-aware spatial prior, thereby improving the discriminability of cross-modal features.

(***iv***) Finally, we visualize the final response maps of our model. As shown in Fig. 11, our model produces clear response maps on several challenging sequences from the FE240 dataset. Even under adverse conditions such as low illumination and fast motion, the response maps consistently exhibit strong uni-modal peaks centered on the target. This indicates high confidence in target localization and further verifies the robustness brought by the synergy between STDV and STVM.

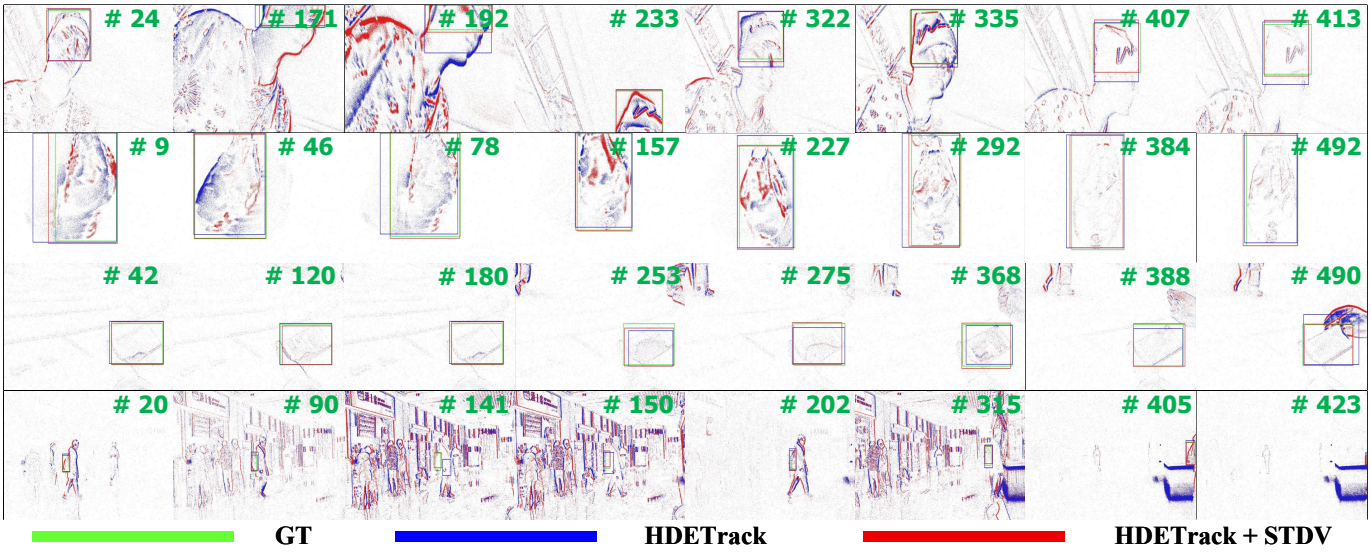


Fig. 9. **Qualitative comparison results** of our method with HDETrack [14] on the EventVOT dataset [14]. The # in the upper-right corner of the images denotes the current frame index.

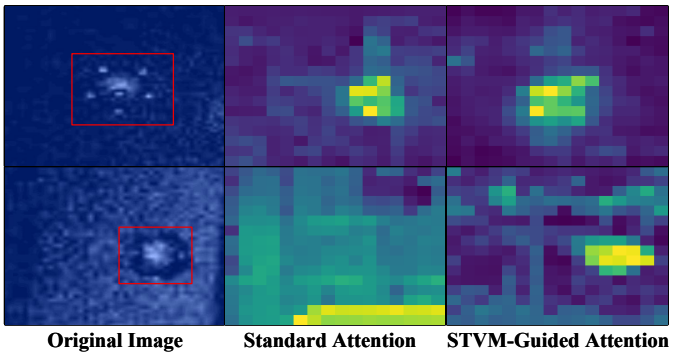


Fig. 10. **Visualization of feature maps** generated by the standard attention and our STVM-guided attention on the FE240 dataset [7].

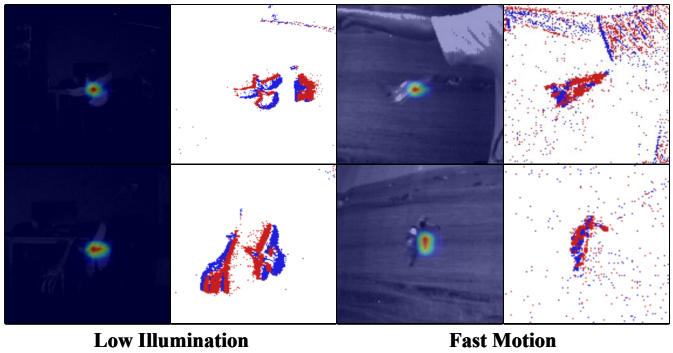


Fig. 11. **Response weight visualization results** of our model on challenging scenes in the FE240 dataset [7]. The final response maps produced by our model feature challenges such as low illumination and fast motion. Our model consistently generates a sharp, well-localized peak on the target, demonstrating its robustness. Event frames are provided for context.

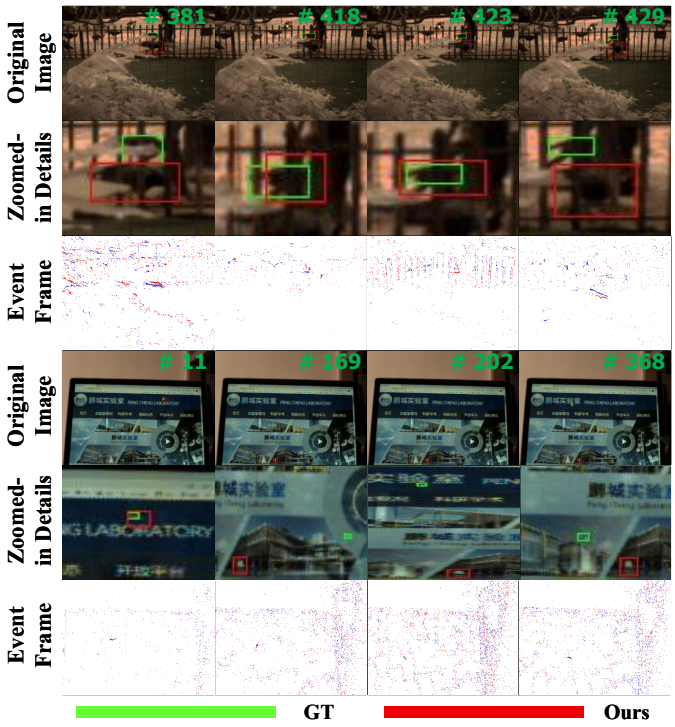


Fig. 12. **Visualization of failure cases** of our method on the COESOT dataset [9]. Event frames are used for visual comparison only.

V. Failure Cases and Limitation Analysis

Although our method effectively exploits spatio-temporal event dynamics for event-based tracking, its advantage may be less pronounced when the target exhibits little or no motion. As illustrated in Fig. 12, when the target remains nearly static or the relative motion between the target and the camera is very weak, only a small number of noisy events are triggered around the target. This insufficient observation leads to weak and incomplete motion information extracted within a limited time window, which is a common problem of existing event representation methods (*e.g.*, as summarized in Table I), thereby rendering the contribution of event-based motion modeling to target tracking insignificant. In particular,

when the quality of RGB images degrades, robust tracking becomes more difficult. This limitation indicates that relying solely on motion-driven event modeling may be ineffective in motion-constrained scenarios; therefore, for sparse event conditions, it is worthwhile to further explore long-term temporal aggregation and event-density-adaptive representation methods.

VI. Conclusion and Future Work

A. Conclusion

In this paper, we have proposed STDV, a learnable and tracking-aware event representation for capturing multi-scale motion dynamics. Through multi-granularity voxelization and hierarchical temporal dilated convolutions, STDV models both short-term motion states and long-term motion trends, providing comprehensive dynamic information tailored to object tracking. Building upon STDV, we have further introduced STVM, which estimates motion intensity by modeling event activity variance across temporal granularities and guides the tracker toward motion-salient regions. Extensive experiments demonstrate that STDV and STVM can serve as effective plug-and-play enhancements for various RGB-event and event-only trackers, consistently improving their performance and establishing new state-of-the-art results. We believe that our work on dynamic event representation offers valuable insights and will inspire future research in dynamic scene perception.

B. Future Work

The results of this work also suggest several promising directions for further exploration. One direction is to develop adaptive temporal aggregation for sparse-event scenarios. In particular, the temporal aggregation range could be dynamically adjusted according to the local event density, allowing the tracker to accumulate sufficient observations when events are sparse while preserving fine temporal resolution during rapid motion. For long-term tracking, the multi-granularity motion cues encoded by STDV could also be incorporated into a confidence-aware template update strategy. Rather than relying solely on motion changes as an update trigger, these cues could be combined with tracking confidence and appearance consistency to identify reliable update moments, thereby reducing template contamination and improving robustness to long-term target variations. Beyond object tracking, the multi-granularity voxelization and hierarchical temporal modeling of STDV may also provide a useful foundation for other event-based vision tasks. With appropriate task-specific prediction heads, learning objectives, and architectural adaptations, STDV could potentially be extended to event-based optical flow estimation, action recognition, and depth estimation.

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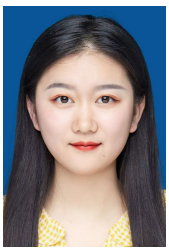
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